

Authorship, Not Delay: What Frontier Distillation Buys an On-Device Small-Language-Model Traffic Controller

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Abstract—Large-language-model traffic signal controllers are evaluated in the cloud; junctions run at the edge, and after an accident someone must prove which controller did what. We present, to our knowledge, the first traffic signal controller fine-tuned and served entirely on-device: a QLoRA-distilled small language model (SLM) compiled to int4 ONNX and served by Microsoft Foundry Local on a 6GB consumer GPU, controlling measured London topologies in SUMO behind a MaxPressure safety shield with a tamper-evident, Ed25519 hash-chained decision audit. Distilling a frontier teacher lifts held-out decision accuracy from 55–69% to 97–99% and saturates by 0.6B parameters: a 3.8B student is statistically indistinguishable from a 0.6B one. Closed-loop, on 30 training-disjoint seeds per topology, this accuracy does *not* translate into significant delay improvement over a fixed-time baseline — but fine-tuning cuts the shield override rate from 0.46–0.60 to 0.11–0.18, raising the SLM’s authored share of served decisions to as much as 0.90, and repairs the one significant harm the stock model causes (grid congestion, +2.6% delay, $p=1.5\times10^{-3}$, to neutral). We argue authorship is the property that matters: an accident audit of a controller whose decisions are mostly shield substitutions attributes nothing to the learned component. Distillation makes the audit *about the model*.

I. INTRODUCTION

LLM-based traffic signal control has moved quickly from prompting to fine-tuning [1]–[3], but the deployment story has not moved with it: every published controller is evaluated against a cloud endpoint, while the signal cabinet at a junction is an edge device with tight memory, no service-level guarantee to the cloud, and — uniquely among LLM application domains — a statutory duty to explain itself after a collision. A junction controller that cannot be audited is not deployable regardless of its delay numbers.

This paper asks what fine-tuning actually contributes to an SLM traffic controller that runs *entirely* on-device under a classical safety shield. We fine-tune with QLoRA [4] on decisions distilled from a frontier teacher, compile to int4 ONNX, and serve with Microsoft Foundry Local on an RTX 2060 (6GB) — consumer hardware of roughly signal-cabinet class. Control runs in SUMO on topologies built from measured London data (Euston A501 with DfT hourly demand, a Bloomsbury grid, Old Street junction), with a MaxPressure [5] shield and anti-starvation floor between the model and the lights, and every decision bound into a hash-chained, signed audit record.

Contributions:

- **A documented QLoRA→int4→Foundry-Local chain.** To our knowledge the first published recipe for serving a custom fine-tuned model through Foundry Local, including the registration and execution-provider steps the toolchain does not document, with bit-identical decision equivalence between CPU and WebGPU serving.
- **A scale-saturation result.** Frontier distillation lifts a 0.6B student to 97–99% holdout agreement with the teacher; a 3.8B student lands on the same numbers. For this task class, capability is not the bottleneck — the distillation ceiling is.
- **The authorship finding.** Offline accuracy decouples from closed-loop delay (neither student significantly beats a fixed-time floor at $n=30$), yet fine-tuning quadruples the fraction of decisions that survive the shield. We introduce authored share as the metric a decision audit needs, and show fine-tuning is what moves it.
- **Honest nulls with mechanism.** We report the closed-loop nulls, the one significant effect (repair of stock-model harm under congestion), and a calibration argument for why the delay headroom was always small.

II. RELATED WORK

LLMLight [1] distils GPT-4 rationales into LoRA-tuned students down to 0.5B and remains the closest prior work; evaluation is on CityFlow grids against cloud-served checkpoints. Traffic-R1 [2] applies R1-style RL to a 3B model; CuraLight [3] is the one fine-tuning work on SUMO; DGLight [6] tunes an 8B model with GRPO and, usefully for calibration, reports the *entire* LLM-vs-MaxPressure gap at 1–2%. iLLM-TSC [7] inverts our topology — an RL agent proposes and a cloud LLM vetoes — which prices the veto path at cloud latency and availability; our deterministic local verifier costs $O(\text{phases})$ per decision. None of these serve on-device, none pair control with a tamper-evident audit, and none evaluate on measured UK topologies. On the safety side, our shield is deliberately weaker than correct-by-construction shielding [8]: it composes preemptive action masking [9], post-posed default substitution, and an anti-starvation runtime monitor, and we discuss what a formal upgrade would require.

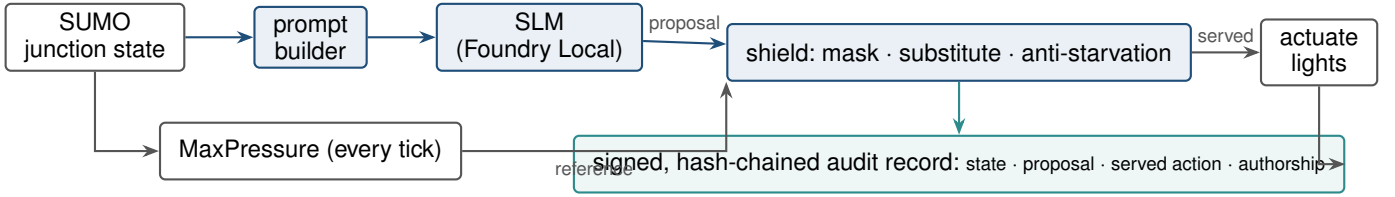


Fig. 1. The junction control loop. The deterministic path (grey) is complete without the SLM; the learned proposer (blue) attaches to it, and every served decision lands in the signed audit chain (teal).

III. SYSTEM AND METHOD

A. Control loop and shield

Each junction is polled on a fixed decision interval (≈ 10 s simulated). The SLM receives junction state in one of four prompt formats of increasing privilege — *myopic* (per-phase halting counts), *sota* (delay-aware), *prediction* (arrival forecasts), *coordination* (neighbour notes) — and must return exactly `{"phase": N}`. The shield masks the menu to conflict-free phases, substitutes the MaxPressure phase when output is invalid, and enforces a minimum-service floor. We define **override rate** as the fraction of served SLM decisions that diverge from what MaxPressure would have chosen and are *not* served (the shield’s choice prevails), and **authored share** as the fraction of all actuations whose served phase originated with the SLM. These are interference metrics in the sense of the shielding literature, not veto counts.

B. Audit

Every decision emits a record (state digest, prompt hash, model proposal, served action, serving authorship, latency) that is Ed25519-signed and hash-chained per junction; chains verify end-to-end after each run. After a simulated collision the system assembles a cited evidence pack against a UK traffic-law knowledge base — deliberately an evidence pack, not a verdict, following the legal-hallucination risk analysis of [10].

C. Distillation and training

We sample 5,952 unique junction states from training-reserved seeds, label them with a frontier teacher (cross-audited against a second frontier model: 98.0–99.0% agreement), and render 59,533 training examples across the four formats with family-correct chat templates and completion-only loss. A hash-based split freezes a 5,580-example holdout before any training. Students: Qwen3-0.6B and Phi-4-mini (3.8B), QLoRA $r=16$ NF4, merged to bf16, compiled to int4 ONNX (RTN, weight-only) with the onnxruntime-genai builder, and registered into Foundry Local. The 0.6B trains on the serving GPU itself in 13.5h; the 3.8B on a single RTX 4090 in 2.3h. The stock control for every comparison is the *same* base model compiled through the *identical* int4 pipeline. CPU and WebGPU execution providers agree bit-identically on holdout decisions (97.42% both), so all sweeps run GPU-served at sub-second p50.

TABLE I

HOLDOUT ACCURACY (%) VS FRONTIER TEACHER, INT4-AS-SERVED. STRICT-JSON COMPLIANCE IN PARENTHESES WHERE BELOW 100%.

	myopic	sota	prediction	coordination
Qwen3-0.6B stock	45.6	55.1 (93.4)	63.3	69.3
Qwen3-0.6B ft	87.9	97.7	97.6	99.0
Phi-4-mini stock	87.5	59.0	86.0	87.0
Phi-4-mini ft	88.7	97.1	97.4	99.3
Phi-4-mini LOTO	88.3	96.8	97.4	99.1

D. Closed-loop protocol

Closed-loop cells pair the controller against a fixed-time baseline on the same seed; evaluation seeds are disjoint from the four training-state seeds. Per topology and arm: $n=30$ seeds, paired t on mean network delay.

IV. EVALUATION

A. Offline: distillation saturates at 0.6B

Table I shows holdout agreement with the frozen teacher labels, stock vs fine-tuned, both students, int4-as-served.

Three observations. First, the two fine-tuned students are statistically indistinguishable despite a $6\times$ parameter gap, and both sit at the teacher-agreement ceiling (the myopic ceiling is intrinsically lower: the teacher was privileged with state the myopic prompt hides). Distillation *saturates* this task at 0.6B. Second, where the gain lands depends on the base model: the 0.6B is lifted everywhere (+32 to +42 points), while Phi-4-mini — already 100% format-compliant and 86–88% accurate on three formats stock — gains substantially only on the delay-aware format (+38.1 points). For a capable base model, fine-tuning buys decision quality, not format discipline. Third, a leave-out-topology student (LOTO, trained with no Old Street states) matches the generalist on every format: the distilled policy does not memorise topology.

B. Closed-loop: delay decouples from accuracy

Table II reports the Qwen3-0.6B closed-loop sweep (fine-tuned ft1 vs identically-compiled stock ft0), paired per-seed against fixed-time.

Neither arm significantly beats fixed-time on any topology; a +42-point offline accuracy gain buys no significant delay change. The one significant closed-loop effect of fine-tuning is protective: stock control significantly *harms* the congested grid (+2.6%, $p=1.5\times 10^{-3}$), the fine-tuned model is neutral there,

TABLE II

CLOSED-LOOP MEAN NETWORK DELAY VS FIXED-TIME BASELINE ($n=30$ DISJOINT SEEDS; PAIRED t). BOLD = SIGNIFICANT.

Topology	Arm	Δ delay	p	override
Euston peak-hour	ft	-3.9%	.16	0.12
	stock	-6.2%	.12	0.47
Bloomsbury grid	ft	+0.7%	.37	0.18
	stock	+2.6%	1.5×10^{-3}	0.46
Old Street	ft	-3.1%	.20	0.11
	stock	-1.1%	.35	0.60
Pooled ($n=90$)	ft	-2.1%	.14	—

and the head-to-head paired difference is -1.8% ($p=.019$). This decoupling has a calibration explanation: with the shield bounding worst-case behaviour for both arms, and the total LLM-vs-MaxPressure headroom reported at 1–2% by work tuning models $13\times$ larger [6], there was never much delay for per-decision accuracy to win. A Phi-4-mini confirmation arm (same 90 cells) lands arm-for-arm on the 0.6B student’s numbers — Euston -3.8% vs -3.9% , Bloomsbury $+0.6\%$ vs $+0.7\%$, override 0.12–0.17, authored share to 0.88 — so the saturation finding holds end-to-end, offline and closed-loop.

Forensics on the failure tail sharpens this into a stronger claim. Of 270 closed-loop cells, 14 are catastrophic ($\geq +20\%$ delay); all their audit chains verify, and the catastrophic seeds are baseline-easy or median realisations, so the failures are a controller property rather than seed hardness. Critically, the two fine-tuned students share every one of their catastrophic (topology, seed) pairs, with per-seed relative delay matching to ± 0.2 points and near-identical decision compositions. Two students of different architecture and $6\times$ different scale, trained separately, fail identically — evidence that distillation converges them onto effectively the same policy. Saturation is not merely a shared accuracy ceiling; it is a shared *policy attractor*.

C. Authorship: what fine-tuning actually moves

The structural effect is unambiguous and consistent across every topology: fine-tuning drops override from 0.46–0.60 to 0.11–0.18 and raises authored share to 0.70–0.90 (stock: 0.54–0.68). The stock controller’s delay numbers are substantially the shield’s delay numbers — most of what it proposes is replaced. The fine-tuned controller *is* the controller. [TODO: Figure: override + authored share, stock vs ft, per topology.]

This is the paper’s claim: for a shielded, audited edge controller, the deployable product of distillation is **authorship**. An accident audit over a stock-model deployment attributes the served decisions to the fallback; the same audit over the distilled deployment attributes them to the model that the operator certified, versioned, and can retrain. Causal accountability — the property regulators and investigators need — is exactly what moved, and it moved by $4\times$.

D. Authorship cuts both ways

If fine-tuning makes the model author the decisions, it also makes the model’s inputs worth attacking. We ran an adversarial trust battery (honest / blatant / stealth neighbour misreporting) against stock and fine-tuned controllers at both scales. Detection is controller-independent by design and in measurement — the ledger scores neighbour claims against local sensing, and a blatant liar is caught and collapsed under every controller (trust ≤ 0.27 , corroboration power stripped) while a calibrated stealth attacker evades everywhere (trust ≥ 0.92). But the delay cost of a blatant attack, relative to each controller’s own honest run, is *larger* for the fine-tuned arms on synthetic grids (+9 to +11 points) than for stock (+3 to +7): a controller that genuinely acts on its coordination inputs is moved further by poisoned ones. The trust ledger is therefore a necessary companion to distillation, not an optional extra — and stealth misreporting, cheap and undetected, is the open problem.

We also measured a specialisation cost. On an adjacent structured task (vehicle-authorisation classification against a curated rule base), the fine-tuned students perform far *worse* than their own stock bases: the 0.6B student cannot emit the task’s `{"class", "grant"}` schema at all without in-prompt scaffolding (26/26 parse failures), having been captured onto `{"phase": N}`. A 200-item MMLU probe shows only mild general-capability decline ($57.5\% \rightarrow 53.0\%$, $p=.37$), so near-neighbour structured tasks are where the cost concentrates. This is an argument for architectural separation, not against distillation: the deterministic gate should keep authorisation, and the distilled model should keep phase selection.

E. Serving reliability at the edge

Sustained serving on consumer hardware is itself a finding: across the campaign we recorded four Foundry Local reliability events (garbage-token emission after long uptime; two unprompted service self-restarts mid-run; registration requiring undocumented synthesis of the inference manifest for builder-compiled models). All were mitigated in harness (endpoint rediscovery, reload-and-retry with intact statistics, fail-loud parse accounting), and p50 decision latency held at 0.4–1.1 s GPU-served — comfortably inside the 10 s decision interval with a $9\times$ margin.

V. LIMITATIONS

Simulation-only, three topologies, one city. Neither student significantly beats the fixed-time floor, and the trust mechanism does not stop a stealth attacker. We distil labels, not reasoning traces — Orca-style trace distillation [11] might transfer more; our myopic format shows the cost of privileged teachers directly. The shield is a heuristic composite, not a synthesized shield with correctness guarantees [8]. “Lossless” int4 is scoped strictly as bit-identical holdout accuracy, weight-only RTN. A seeded 200-item MMLU retention probe through the same endpoints finds no significant general-capability loss at 3.8B (57.5% stock vs 53.0% fine-tuned, $p=.37$); at 0.6B

the stock model already scores at chance on this format, so forgetting is unmeasurable there by floor effect.

VI. CONCLUSION

A 0.6B SLM, distilled from a frontier teacher for 13 GPU-hours on the same 6GB device that serves it, reaches the teacher-agreement ceiling; six times the parameters add nothing. Closed-loop it does not beat a fixed-time floor — and neither does anything else within the shield’s envelope — but it stops needing the shield: override falls $4\times$, authored share reaches 0.90, and the stock model’s one significant harm disappears. For edge traffic control, where every actuation must survive an audit, that conversion — from shield-driven to model-authored control — is the contribution fine-tuning makes. Future work: GRPO-style closed-loop tuning [6], formal shield synthesis, and a hardware-in-the-loop pilot.

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